

Assignment 4

00103335: Deep Learning and Reinforcement Learning

Fall 2026

Sources. Problems 1–3 are adapted from M. J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Exercises 2.7, 2.8, and 2.12. Problem 5 is adapted from R. Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*, Corollary 4.2.13. Problem 4 is an additional exercise on truncation.

Problem 1. Bennett’s inequality

Let $X_1, \dots, X_n$ be independent, mean-zero random variables with $|X_i| \leq b$ almost surely, where $b > 0$. Write $\sigma_i^2 = \mathbb{E}X_i^2$ and $v = n^{-1} \sum_{i=1}^n \sigma_i^2 > 0$. Define $h(u) = (1+u) \log(1+u) - u$ for $u \geq 0$.

- (a) Show that, for every $\lambda \geq 0$,

$$\log \mathbb{E} e^{\lambda X_i} \leq \frac{\sigma_i^2}{b^2} (e^{\lambda b} - 1 - \lambda b).$$

- (b) Prove that, for every $t > 0$,

$$\mathbb{P} \left(\frac{1}{n} \sum_{i=1}^n X_i \geq t \right) \leq \exp \left[-\frac{nv}{b^2} h \left(\frac{bt}{v} \right) \right].$$

State the corresponding lower-tail bound.

- (c) Prove that $h(u) \geq u^2/[2(1+u/3)]$ for $u \geq 0$, and deduce the two-sided Bernstein bound

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n X_i \right| \geq t \right) \leq 2 \exp \left[-\frac{nt^2}{2(v+bt/3)} \right].$$

Problem 2. Bernstein tails and expected error

Let $Z \geq 0$ satisfy, for all $t \geq 0$,

$$\mathbb{P}(Z \geq t) \leq C \exp \left[-\frac{t^2}{2(\nu^2 + bt)} \right], \quad C \geq 1, \quad \nu, b > 0.$$

- (a) Prove

$$\mathbb{E}Z \leq 2\nu(\sqrt{\pi} + \sqrt{\log C}) + 4b(1 + \log C).$$

- (b) Let $X_1, \dots, X_n$ be i.i.d. mean-zero variables with $\sigma^2 = \mathbb{E}X_1^2$, and write $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Suppose they satisfy the Bernstein moment condition

$$|\mathbb{E}X_1^k| \leq \frac{k!}{2} \sigma^2 b^{k-2}, \quad k = 2, 3, \dots$$

Show that their empirical mean obeys

$$\mathbb{E}|\bar{X}_n| \leq \frac{2\sigma}{\sqrt{n}} (\sqrt{\pi} + \sqrt{\log 2}) + \frac{4b}{n} (1 + \log 2).$$

Problem 3. Maxima of light-tailed variables

Let $X_1, \dots, X_n$ be mean-zero random variables. No independence assumption is made.

- (a) Suppose each X_i is sub-Gaussian with the same variance proxy σ^2 , where $\sigma \geq 0$: $\mathbb{E}e^{\lambda X_i} \leq e^{\sigma^2 \lambda^2 / 2}$ for all $\lambda \in \mathbb{R}$. Prove, for $n \geq 1$,

$$\mathbb{E} \max_{1 \leq i \leq n} X_i \leq \sigma \sqrt{2 \log n}.$$

- (b) Under the same assumption, prove, for $n \geq 2$,

$$\mathbb{E} \max_{1 \leq i \leq n} |X_i| \leq \sigma \sqrt{2 \log(2n)} \leq 2\sigma \sqrt{\log n}.$$

- (c) Now replace the global MGF assumption by the sub-exponential condition

$$\mathbb{E}e^{\lambda X_i} \leq e^{\nu^2 \lambda^2 / 2} \quad \text{for } |\lambda| < 1/b,$$

with common $\nu, b > 0$. Prove, for $n \geq 2$,

$$\mathbb{E} \max_{1 \leq i \leq n} X_i \leq \nu \sqrt{2 \log n} + b \log n, \quad \mathbb{E} \max_{1 \leq i \leq n} |X_i| \leq \nu \sqrt{2 \log(2n)} + b \log(2n).$$

Problem 4. Symmetric Pareto variables and truncation

Let $X_1, \dots, X_n$ be i.i.d. with density

$$f_\alpha(x) = \frac{\alpha}{2} |x|^{-\alpha-1} \mathbf{1}_{\{|x| \geq 1\}}, \quad \alpha > 2.$$

- (a) Show that $\mathbb{P}(|X_i| > u) = u^{-\alpha}$ for $u \geq 1$, $\mathbb{E}X_i = 0$, and $\sigma_\alpha^2 := \mathbb{E}X_i^2 = \alpha/(\alpha-2)$.
 (b) For a threshold $u \geq 1$, set $Y_i = X_i \mathbf{1}_{\{|X_i| \leq u\}}$. Show that $\mathbb{E}Y_i = 0$, $|Y_i| \leq u$, and $\mathbb{E}Y_i^2 \leq \sigma_\alpha^2$.
 (c) Prove the concentration bound

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left[-\frac{t^2}{2(n\sigma_\alpha^2 + ut/3)}\right] + nu^{-\alpha} \quad (t > 0, u \geq 1).$$

- (d) Fix $\delta \in (0, 1)$. Choose the smallest $u \geq 1$ for which the truncation term $nu^{-\alpha}$ in part (c) is at most $\delta/2$. Set $t = n\varepsilon$, and solve for the positive $\varepsilon = \varepsilon_{n,\delta}$ that makes the exponential term in part (c) equal to $\delta/2$. Give u and $\varepsilon_{n,\delta}$ explicitly, and conclude that

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq n\varepsilon_{n,\delta}\right) \leq \delta.$$

For fixed δ , determine the leading order of $\varepsilon_{n,\delta}$ as $n \rightarrow \infty$.

Problem 5. Covering the Euclidean ball

Let $d \geq 1$ be an integer. Write $B_2^d = \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$ and $S^{d-1} = \{x \in \mathbb{R}^d : \|x\|_2 = 1\}$. For $K \subseteq \mathbb{R}^d$, let $N(K, \varepsilon)$ be the smallest size of an ε -net *with centers in K* under Euclidean distance.

- (a) Prove $N(B_2^d, \varepsilon) \geq \varepsilon^{-d}$.
 (b) Prove, for every $\varepsilon > 0$,

$$N(B_2^d, \varepsilon) \leq (1 + 2/\varepsilon)^d.$$

- (c) Prove that $N(S^{d-1}, \varepsilon) \leq (1 + 2/\varepsilon)^d$, with net points on the sphere. In particular, give the bound obtained when $\varepsilon = 1/4$.